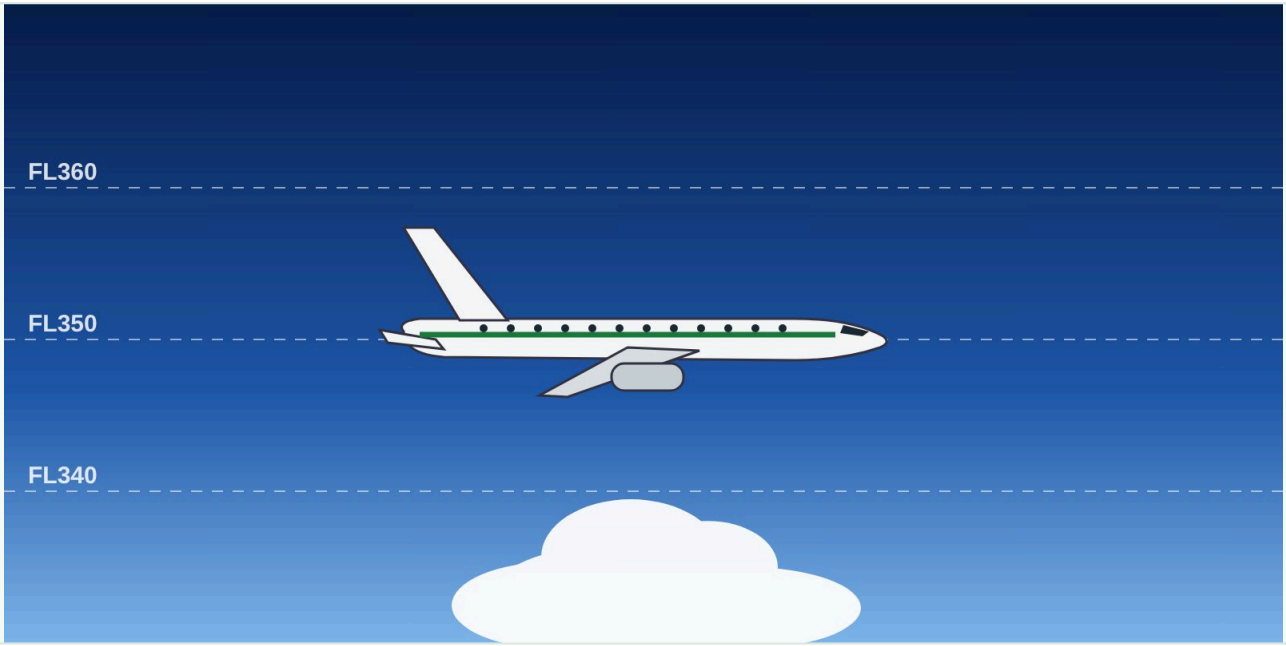
MODULE 1 · RVSM OPERATIONS

Lesson 03: Entering and Flying in RVSM Airspace

Lesson 3 of 6 · about 7 min

LEARNING OBJECTIVES

- List the equipment that must be operating before entering RVSM airspace.
- Apply the en-route height-keeping procedures.
- Perform and record the altimeter cross-checks.



Fly the cleared flight level with the autopilot altitude hold engaged.

Before entering RVSM airspace

The following equipment must be operating normally:

- two primary altitude measurement systems;
- one automatic altitude control system;
- one altitude alerting device;
- the transponder.

If any of this equipment fails before entry, **request a new clearance** to avoid RVSM airspace, or follow the established contingency procedures.

En-route procedures

- Fly at the **cleared flight level (CFL)**. Make sure every ATC clearance is fully understood, read back and cross-checked by both pilots.
- Except in an emergency, do not intentionally leave the CFL without an ATC clearance.
- Use the **automatic altitude capture** for level changes. Overshoots and undershoots should not exceed **150 ft**.

- Reduce the vertical rate to **less than 1,500 ft/min in the last 1,000 ft** before the cleared level, especially when traffic is at the adjacent level (see Lesson 5).
- In cruise, fly with the **automatic altitude control engaged**, except when the aircraft must be re-trimmed or in turbulence where the procedure requires otherwise.

Altimeter cross-checks

- At intervals of about **one hour**, cross-check the primary and standby altimeters.
- The two primary altimeters must agree within **200 ft**.
- **Record** at least the initial cross-check — on the operational flight plan (Jetplan).
- Normally, the altimetry system controlling the autopilot should also be the source for the transponder's altitude reporting. If not, report the system as defective and notify ATC.

KEY TAKEAWAYS

- Before entry: two altimetry systems, autopilot altitude hold, altitude alerter, transponder.
- Fly the CFL with altitude hold engaged; overshoot or undershoot less than 150 ft.
- Less than 1,500 ft/min in the last 1,000 ft before the cleared level.
- About every hour: primary altimeters within 200 ft; record the initial check.

KNOWLEDGE CHECK

What is the maximum difference allowed between the two primary altimeters in RVSM airspace?

A. 75 ft
B. 150 ft
C. 200 ft
D. 300 ft

Answer: C. The primary altimeters must agree within 200 ft.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.